

HEIDRUN, Norway

WMO: 011025

Lat: 65.333N

Lon: 7.317E

Elev: 262

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.8	28.7	14.1	11.4	30.7	16.2	12.6	31.3	66.7	41.6	62.0	41.5	25.2	100	1.123

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	64.1	58.9	61.2	57.3	59.4	56.0	59.6	62.8	58.1	61.0	56.8	59.5	18.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.6	71.5	60.7	56.8	69.5	59.8	55.4	66.0	58.2	26.2	62.5	25.3	60.3	24.3	59.5	64.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 58.7	52.1	47.0	24.0	67.8	2.6	3.4	22.2	70.2	20.7	72.2	19.2	74.1	17.4	76.5		
(5)			21.8	61.4	2.1	2.1	20.2	62.9	19.0	64.1	17.8	65.3	16.2	66.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.1	38.8	38.6	38.7	41.8	45.1	49.0	54.0	54.9	51.8	46.4	42.0	40.1
	DBStd	7.22	4.56	4.57	4.54	4.44	4.43	3.91	4.17	3.42	3.55	4.26	4.69	4.56
	HDD50	2190	348	319	351	249	161	65	6	2	21	123	239	306
	HDD65	7248	813	739	816	697	616	480	342	313	396	576	689	771
	CDD50	419	0	0	0	2	11	35	129	154	76	12	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	23.6	28.3	27.0	26.1	22.6	19.6	18.6	18.3	19.2	23.1	25.3	26.5	28.5
	PrecAvg	40.2	5.2	3.9	3.2	2.2	1.5	1.9	2.4	3.3	3.8	4.2	4.0	4.6
(15) Precipitation	PrecMax	48.7	10.5	6.9	5.6	4.3	2.6	3.9	3.5	5.0	5.7	7.5	6.4	7.9
	PrecMin	25.5	2.4	1.1	1.0	1.0	0.8	0.7	1.1	1.7	1.8	1.5	0.8	2.0
	PrecStd	5.7	1.9	1.6	1.3	0.8	0.4	0.7	0.6	1.0	1.2	1.7	1.5	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	46.8	48.4	55.0	60.8	64.0	67.9	66.2	60.9	56.9	51.8	48.6
		MCWB	46.0	44.4	44.3	47.9	52.9	56.0	60.5	59.6	56.3	52.6	49.3	46.6
	2%	DB	46.7	46.5	46.6	51.5	55.8	59.1	64.5	63.0	59.1	54.0	50.2	48.2
		MCWB	44.6	44.3	43.9	46.4	50.3	54.2	59.5	58.7	55.4	50.8	48.2	46.6
	5%	DB	46.2	45.0	46.0	48.4	53.4	56.8	62.3	61.1	57.4	53.2	48.6	46.7
		MCWB	44.1	42.6	43.5	45.5	48.9	52.6	58.3	57.7	54.5	50.3	46.8	44.8
	10%	DB	44.8	44.7	44.7	46.7	50.3	54.0	59.4	59.3	55.8	51.8	48.1	46.3
		MCWB	42.5	42.2	42.6	44.6	47.4	50.9	56.2	56.4	53.4	49.2	46.3	44.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.5	45.9	45.8	48.7	52.8	57.1	61.9	61.2	57.0	53.2	50.3	48.0
		MCDB	46.9	46.3	46.7	52.4	58.8	61.9	65.8	64.8	59.8	55.5	50.9	48.1
	2%	WB	45.1	44.6	44.7	46.9	51.1	54.8	60.0	59.6	55.9	52.0	48.6	46.6
		MCDB	45.9	45.5	45.5	49.3	55.3	58.5	63.7	62.4	58.2	54.0	49.3	47.2
	5%	WB	43.9	43.4	43.6	45.9	49.5	53.1	58.5	58.4	55.3	50.9	47.4	45.3
		MCDB	45.2	44.8	45.0	47.5	52.3	55.8	61.8	60.9	57.4	52.6	48.5	46.3
	10%	WB	42.5	42.1	42.4	44.9	48.0	51.8	56.9	57.2	54.5	49.8	46.0	43.8
		MCDB	44.4	44.0	44.3	46.3	50.2	54.4	59.8	59.6	56.5	51.3	47.6	45.5
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	5.4	5.4	5.1	5.0	4.3	4.5	4.4	4.4	4.2	5.1	5.4
		MCDBR	5.5	5.3	5.7	6.6	8.2	7.9	8.8	6.8	5.7	4.7	5.0	5.9
		MCWBR	6.6	6.0	6.1	5.0	5.1	5.2	5.3	5.0	4.4	4.6	5.5	6.7
	5% WB	MCDBR	5.5	5.5	5.7	5.7	7.5	7.2	8.3	6.3	5.3	4.5	5.0	5.8
		MCWBR	7.0	6.6	6.9	5.1	5.2	5.2	5.3	5.2	4.8	5.1	6.1	7.4
(40) Clear-Sky Solar Irradiance	taub	0.374	0.325	0.341	0.361	0.350	0.349	0.357	0.351	0.340	0.320	0.357	0.372	
	taud	2.392	2.185	2.306	2.347	2.456	2.497	2.496	2.507	2.488	2.297	2.426	2.362	
	Ebn at Noon	65	169	230	255	272	275	268	259	234	172	67	16	
	Edh at Noon	8	18	25	30	30	30	29	26	21	16	7	3	
(44) All-Sky Solar Radiation	RadAvg	25	140	496	1033	1524	1589	1428	1092	653	246	47	7	
	RadStd	2	17	47	96	80	176	158	102	77	36	6	0	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page